

Day 36 - Lagrangian Examples III

WHEN YOU START TO LEARN LAGRANGIAN MECHANICS



Announcements

- Homework 8 is "Late" 24 Apr
 - Last Exercise 0: Reflect Learning Outcomes
- Final Project is posted
 - Video Presentations due 27 Apr
 - Computational Essay due 1 May
 - Rubric for both are posted
- No class (20 Apr - 24 Apr) - DC out of country
 - Make appointment for project help (clicker extra credit)

Announcements

Rest of Semester Schedule

- CW16 - Examples of Lagrangian Dynamics (HW8)
- CW17 - Project Prep (DC out of country)
- CW18 - Final Project Due
 - Video Presentations due 27 Apr
 - Computational Essay due 1 May

NO IN-CLASS FINAL EXAM

Reminders

We found the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

where M is the mass of the left block, m is the mass of the right block, M_p is the mass of the pulley, R is the radius of the pulley, and ϕ is the angle of rotation of the pulley.

We used the **scleronomic** constraint $\dot{y}_1 = R\dot{\phi}$ to do this.

Holonomic constraints are those that can be expressed as a function of the coordinates and **time**, $f(q_1, q_2, \dots, q_n, t) = 0$.

Clicker Question 36-1a

We derived the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

What is generalized force, $F_\phi = \partial\mathcal{L}/\partial\phi$?

1. $+(M - m)gR$
2. $-(M - m)gR$
3. $+(M + m)R^2\dot{\phi}$
4. $-(M + m)R^2\dot{\phi}$
5. Something else

Clicker Question 36-1b

For the Atwood's machine with a rotating pulley, what are the units of the generalized force, F_ϕ ?

1. Newtons
2. Joules
3. Newton-meters
4. Joule-seconds
5. Something else

Clicker Question 36-2a

We derived the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

What is the generalized momentum, $p_\phi = \partial\mathcal{L}/\partial\dot{\phi}$?

1. $+(M - m)gR$
2. $-(M - m)gR$
3. $+(M + m)R^2\dot{\phi}$
4. $-(M + m)R^2\dot{\phi}$
5. Something else

Clicker Question 36-2b

For the Atwood's machine with a rotating pulley, what are the units of the generalized momentum, p_ϕ ?

1. Newtons
2. Joules
3. Newton-meters
4. Joule-seconds
5. Something else

Clicker Question 36-3a

Two blocks (m) connected by a spring (k, L) on a horizontal frictionless surface. The first block is described by a coordinate x_1 and the second block is described by a coordinate x_2 from a mutual origin. What is the Lagrangian for this system?

1. $\mathcal{L} = T - U = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2 - L)^2$

2. $\mathcal{L} = T - U = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2)^2$

3. $\mathcal{L} = T - U = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2 + L)^2$

4. $\mathcal{L} = T - U = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 + x_2 - L)^2$

5. Something else

Clicker Question 36-3b

For the two block and spring system, what is the equation of motion for the first block?

1. $m\ddot{x}_1 = -k(x_1 - x_2)$
2. $m\ddot{x}_1 = -k(x_1 - x_2 + L)$
3. $m\ddot{x}_1 = -k(x_1 - x_2 - L)$
4. $m\ddot{x}_1 = -k(x_1 + x_2 - L)$
5. Something else

Clicker Question 36-3c

For the two block and spring system, what is the equation of motion for the second block?

1. $m\ddot{x}_2 = +k(x_1 - x_2)$
2. $m\ddot{x}_2 = -k(x_1 - x_2 - L)$
3. $m\ddot{x}_2 = +k(x_1 - x_2 - L)$
4. $m\ddot{x}_2 = +k(x_1 + x_2 - L)$
5. Something else

Clicker Question 36-4a

Sometimes, we can choose better or more useful generalized coordinates. For the two block and spring system, we can choose the center of mass coordinate, $x_{cm} = (x_1 + x_2)/2$, and the relative coordinate, $x = x_1 - x_2$. What is the Lagrangian for this system in these new coordinates?

1. $\mathcal{L} = T - U = m\dot{x}_{cm}^2 + \frac{1}{4}m\dot{x}^2 - \frac{1}{2}k(x - L)^2$
2. $\mathcal{L} = T - U = m\dot{x}_{cm}^2 + \frac{1}{4}m\dot{x}^2 - \frac{1}{2}k(x + L)^2$
3. $\mathcal{L} = T - U = m\dot{x}_{cm}^2 + \frac{1}{4}m\dot{x}^2 - \frac{1}{2}k(x)^2$
4. $\mathcal{L} = T - U = m\dot{x}_{cm}^2 + \frac{1}{4}m\dot{x}^2 - \frac{1}{2}k(x + L)^2$
5. Something else

Clicker Question 36-4b

For this new Lagrangian, what is the equation of motion for the relative coordinate, x ?

1. $\frac{1}{2}m\ddot{x} = -k(x - L)$
2. $\frac{1}{2}m\ddot{x} = -k(x + L)$
3. $\frac{1}{2}m\ddot{x} = -kx$
4. $\frac{1}{2}m\ddot{x} = -k(x + L)$
5. Something else

What is the equation of motion for the center of mass coordinate, x_{cm} ?

Clicker Question 36-5a

Consider a bead sliding on a parabolic bowl described by the constraint $z = c\rho^2$ where ρ is the distance from the vertical axis. The Lagrangian for this system in Cartesian coordinates is:

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Don't use the constraint, what are the equations of motion for this system? Do they seem correct?

Click anything to indicate you are ready to see the answer.

Clicker Question 36-5b

For the constraint for the bead in a parabolic bowl ($z = c\rho^2$), what are the units of c ?

1. $[L^2]$
2. $[L^{-2}]$
3. $[L]$
4. $[L^{-1}]$
5. Something else

Clicker Question 36-5c

Now use the constraint to write the Lagrangian for the bead in a parabolic bowl in cylindrical coordinates, (ρ, ϕ, z) . What is the Lagrangian for this system?

1. $\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2) - mgc\rho^2$
2. $\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2) - mgc\rho^2$
3. $\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2\dot{\rho}^4) - mgc\rho^2$
4. $\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2\dot{\rho}^2) - mgc\rho^2$
5. Something else

Hint: $v^2(\rho, \phi, z) = \dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2$

Clicker Question 36-5d

For the bead in a parabolic bowl, there is a generic Lagrangian:

$$\mathcal{L}(\rho, \dot{\rho}, \phi, \dot{\phi}, z, \dot{z}, t)$$

How many coordinates are there, truly? **here, each variable is a coordinate**

- A. 2
- B. 3
- C. 4
- D. 5
- E. None of these

Which coordinates are independent?

Clicker Question 36-5e

The Lagrangian for the bead in a parabola does not depend on which of the following?

1. ρ
2. ϕ
3. z
4. More than one of these
5. None of these

When a coordinate does not appear in the Lagrangian, it is called a **cyclic** or **ignorable** coordinate. This means that the generalized momentum associated with that coordinate is conserved.